GEOGRAPHICAL INFORMATION SYSTEM (GIS) APPLICATION ON A DESCRIPTIVE SOIL COMPOSITION ANALYSIS IN NASSAU COUNTY, NEW YORK

Miranda Maliszka[1], Anthony Johnson[1], and Antonios Marsellos[1]

[1]*Dept. Geology, Environment and Sustainability, Hofstra University, NY, USA*

Topic Introduction

- Long Island topography is a direct product of the Wisconsin Glacial Episode that occurred approximately 22,000 years ago
- Harbor Hill and Ronkonkoma moraine with bi-products consist of glacial till which can be classified as Acrisol
- Acrisol is primarily made-up of boulders, gravel, sand, silt and clay
- The outwash plain comprises stream deposits from the glacial melt water that include well sorted sand and gravel
- Over time stream deposits along the south shoreline have been refined by waves and ocean currents to form barrier beaches.

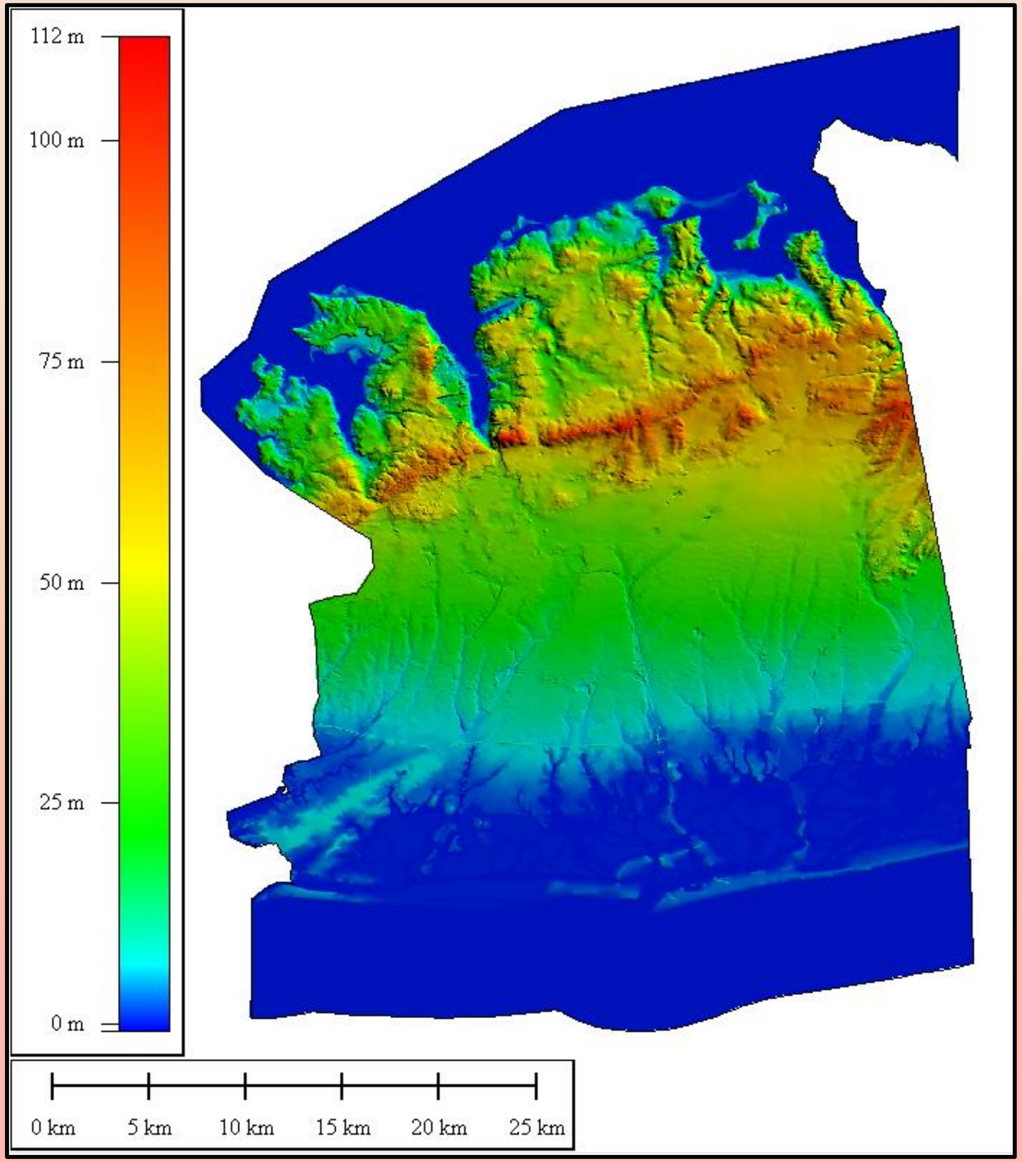

Figure 1: Digital Elevation Model (DEM) of Nassau county in Long Island, NY created in Global Mapper using a 10-meters resolution from the National Elevation Dataset (NED) shows the current topography.

Our Research

- Purpose
  - Comparison analysis between the most recent soil data dating back to the early 1900s
- Current Day (2020) Soil Data
  - The United States Department of Agriculture
  - Natural Resources Conservation Service
  - Food and Agriculture Organization of the UN
- Historical (1903) Nassau County
  - Field Operations of the Bureau of Soils
  - Hempstead Loam/Hempstead Gravelly Loam
    - High percentages of silt
  - Galveston Sandy Loam
    - High percentages of sand
  - Galveston Clay
    - Mud and Gravel

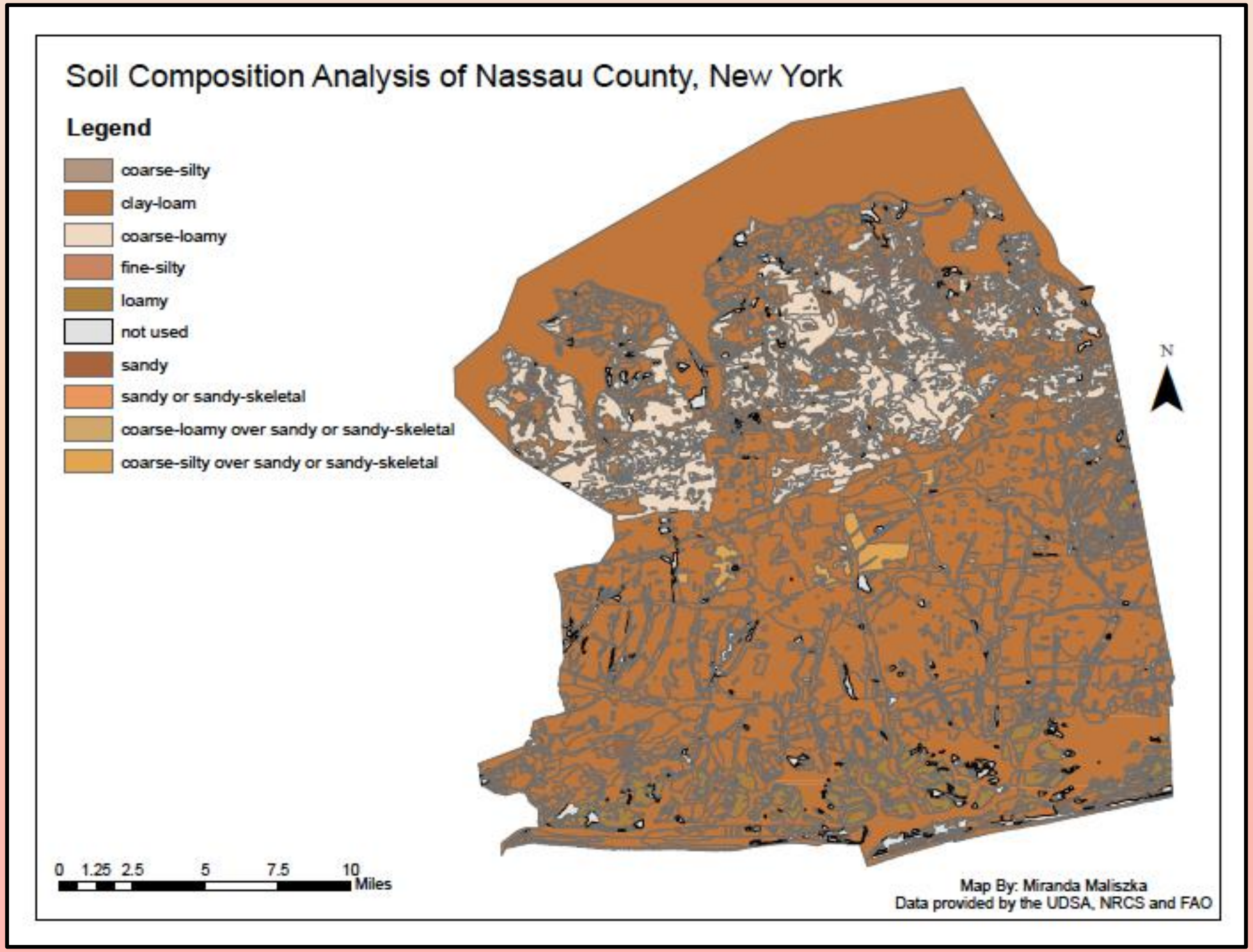

Figure 2: Map created by this research showcasing the soil composition of Nassau County, New York using data provided by the United States Department of Agriculture, Natural Resources Conservation Service

Methodology

- SSURGO database was used to run the Nassau County tabular data
- ArcMap used to process and project the spatial data which excluded soil text classification as one of the attributes
- Unsurveyed areas were identified using a Digital Soil Map of the World provided by the FAO of the United Nations
  - Soil classification of Dystric Cambisols (Bd)
  - Composition: 32.7% sand, 30.3% silt, 37.1% clay
  - Soil triangle identified it as Clay-Loam
- Earliest record of soil surveys for Nassau County, New York was 1903, by The Field Operations of the Bureau of Soil

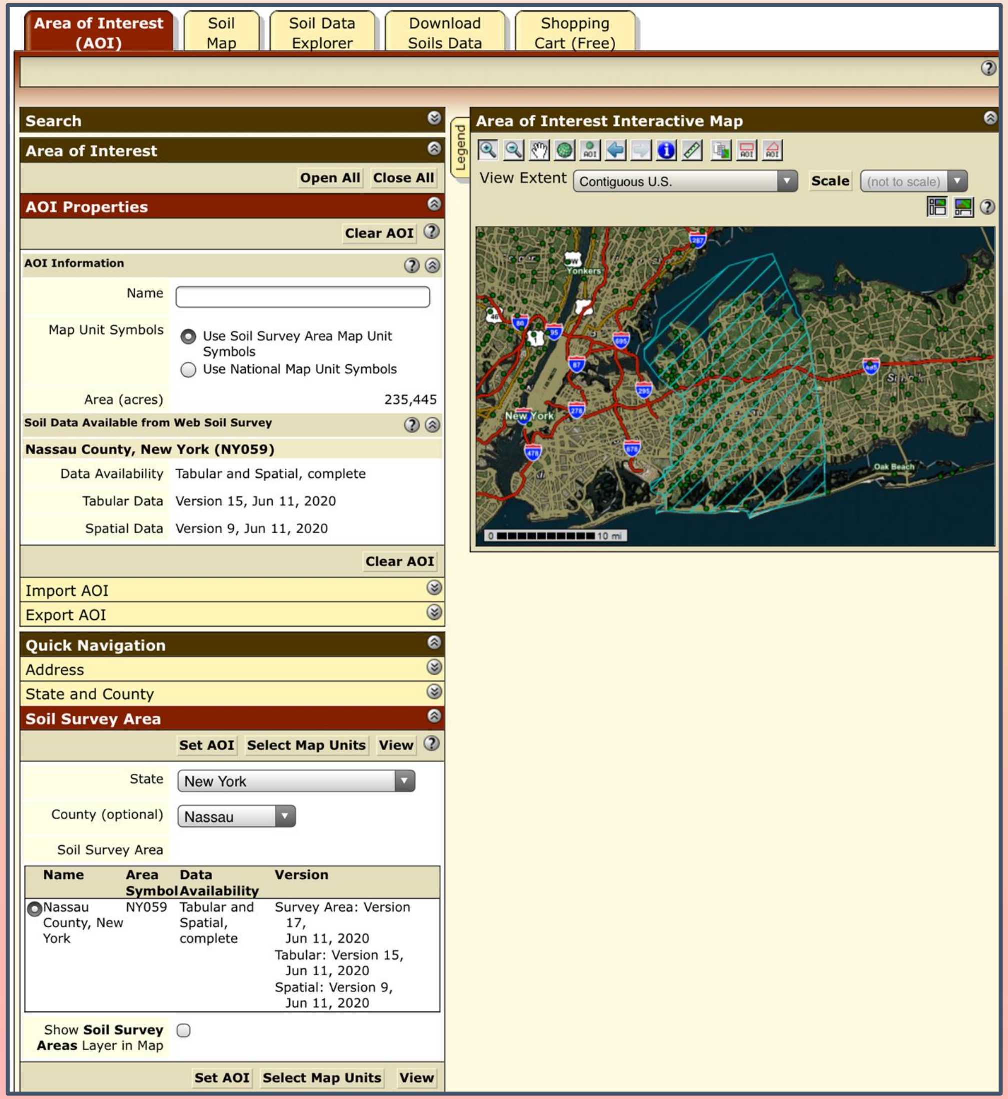

Figure 3: Photo of the USDA, NRCS SSURGO Database that houses the 2020 Soil Survey used to produce the map shown as figure two.

2020 Soil Survey Data Results

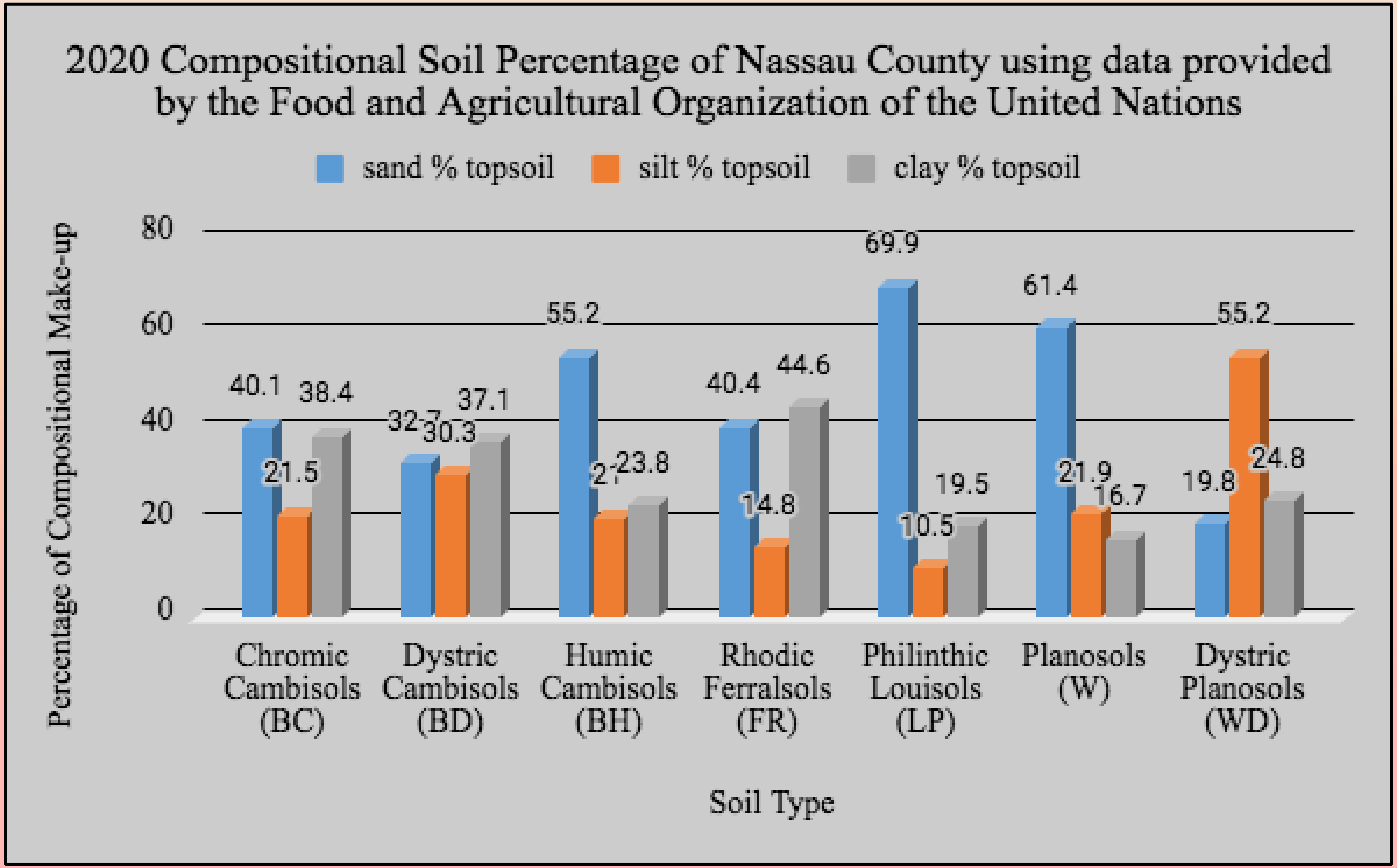

Figure 3: Bar Graph representing the topsoil percentage breakdown of sand, silt and loam in order to identify the most common soils in Nassau County, New York by using data provided by the Food and Agricultural Organization of the United Nations (2020).

1903 Soil Survey Data Results

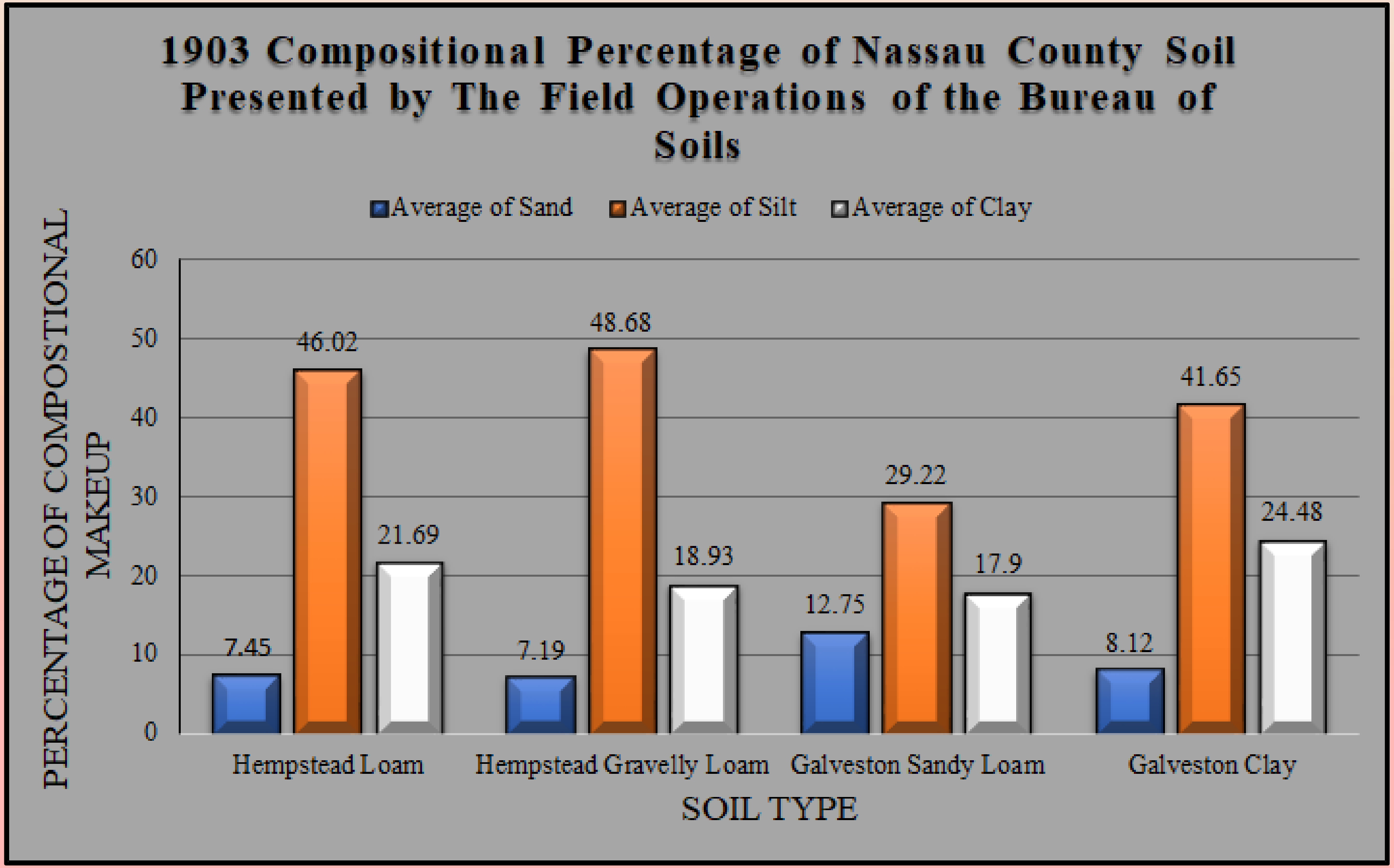

Figure 4: Bar Graph representing the soil make-up percentages of sand, silt and clay using data provided by The Field Operations of the Bureau of Soils (1903).

Discussion

- Substantial difference between the percentages of sand, silt, and clay overall
- 1903 survey data, there is a strong dominance of silt between all four samples while today sand is more prominent amongst all seven samples
- Environmental factors to consider
    - Greenhouse gas emission
    - Increase in Precipitation rates
    - Erosion
    - Poor Soil Health

Conclusion

- Map was produced to visually represent the current soil types located in Nassau County, New York

- In addition, 1903 soil survey data provided by the Field Operations of the Bureau of Soils was used further to compare the progression of soil over the past century

- With that knowledge in conjunction with current environmental factors such as climate change and erosion, a better understanding of how the soil composition has changed over time was cultivated

- In conclusion, over the past hundred and fifteen years a county once dominating in silt has slowly been transformed to a ground of higher percentages of sand and clay

Acknowledgements

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Resources

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